Artificial Intelligence II | MiBA
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Classification Assignment

Predicting loan default

LendingClub

Goal Definition

Develop a model to **manage risk proactively after loan issuance** by **predicting default risk before a borrower exhibits clear signs of financial distress** or requires intervention (such as hardship plans or debt settlement).

This is particularly relevant in the secondary loan market, where investors purchase already-issued loans and need to assess their ongoing risk. The model's predictions will be used to evaluate investment strategies, optimize loan selection, and quantify risk exposure by applying default probabilities to open loans. In a real-world scenario, this approach enables early risk detection and facilitates preemptive strategies to minimize defaults, enhance portfolio performance, and maximize returns.

Business setting: This project focuses on the secondary market of Lending Club, where investors can purchase already-issued loans. From the perspective of an investor who has not yet invested, the goal is to assess and optimize loan selection before committing capital. By leveraging historical data on closed loans, a predictive model is built to estimate default probabilities for open loans. This allows the investor to evaluate different investment strategies based on credit scores, interest rates, and risk exposure, ultimately enhancing decision-making and maximizing returns."**

Expected Outcomes

To assess the effectiveness and business implications of the predictive model, we will:

1) **Compare the confusion matrices** of different models to evaluate their classification performance.
2) **Analyze the business impact** of false positives (**FP**), false negatives (**FN**), true positives (**TP**), and true negatives (**TN**), ensuring that the cost of misclassification is considered.
3) **Modify the decision threshold** to balance cost sensitivity effectively, considering trade-offs between precision and recall.
4) **Determine business implications** of the best-performing model based on confusion matrix results, assessing its practical usability in real-world loan risk management.
5) **Apply the model** trained on “closed” loans (fully paid or defaulted) **to “open” loans** (ongoing repayments) to estimate expected business impact.
6) **Compare different investment strategies across various credit grades**, evaluating profitability based on default rates, interest returns, and risk exposure. The goal is to determine the **optimal strategy to maximize returns while effectively managing risk.**

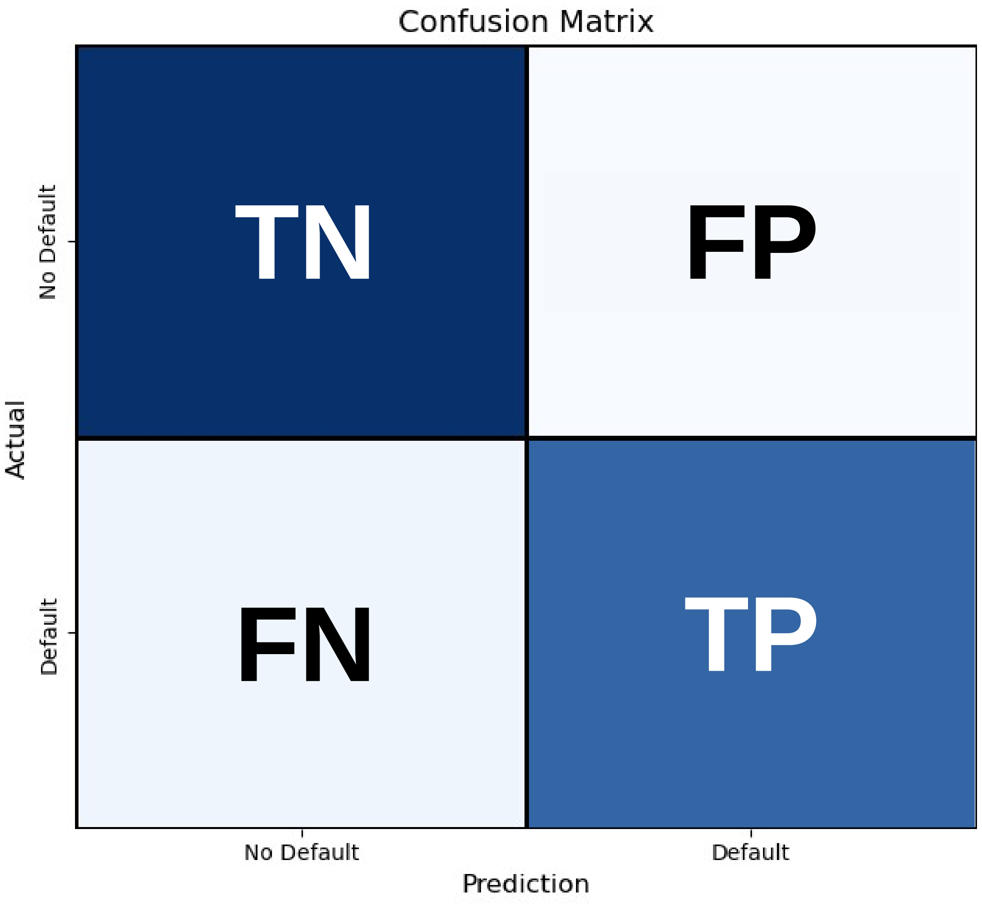
Preprocessing of the data

Key transformations made in the preprocessign of the closed loans dataset...

1) A qualitative analysis of input features was performed to identify and exclude variables that could introduce data leakage.
2) The target variable, loan_status, was transformed into a binary classification label: 1 indicating default, and 0 indicating no default.
3) Feature engineering introduced six new variables derived from existing features, capturing time elapsed since key events and transformations of FICO scores to enhance model signal.
4) Outliers were addressed using the Interquartile Range (IQR) method.
5) Missing values were imputed using two different strategies depending on their underlying mechanism—distinguishing between missing completely at random (MCAR) and missing not at random (MNAR).
6) Categorical features were encoded using a combination of ordinal encoding and one-hot encoding, as appropriate.
7) Numerical features were standardized using a StandardScaler to normalize their distributions.
8) Feature selection was performed using an automated approach based on both pairwise correlation and mutual information with the target. When two features exhibited high correlation, the one with lower mutual information was discarded to reduce multicollinearity and retain predictive relevance.

All preprocessing steps were encapsulated within a scikit-learn **Pipeline**, enabling robust integration for model training, evaluation on new data, and experimentation with alternative algorithms—while ensuring that **no data leakage** occurs during cross-validation or deployment.

Business impact of classification outcomes



True Negative (TN): The model correctly identifies a loan as low risk, and the borrower continues making payments without defaulting. This leads to a stable investment with predictable returns. The investor avoids unnecessary exclusion of good loans, ensuring a balanced portfolio with sustainable profits.

True Positive (TP): The model correctly predicts a high-risk loan, which subsequently defaults. This prevents an investment in a bad loan, avoiding potential losses. A high TP rate improves risk management by ensuring that investors avoid loans likely to default, thereby optimizing capital allocation.

False Negative (FN): The model incorrectly classifies a high-risk loan as low-risk, leading to an investment in a loan that later defaults. This results in direct financial loss, reduced returns, and higher exposure to risk. FN errors are particularly costly in the secondary market, as they could lead to mispricing of loans, causing the investor to overpay for a loan that has a high likelihood of defaulting.

False Positive (FP): The model incorrectly classifies a low-risk loan as high-risk, leading to missed investment opportunities. While this prevents potential exposure to bad loans, it also means rejecting loans that could have provided stable interest income. A high FP rate could cause overly conservative investment strategies, reducing overall returns.

Tradeoff between FN and FP

False Negatives ⟸ trade-off between prioritizing optimization of FN vs. FP ⟹ **False Positives**

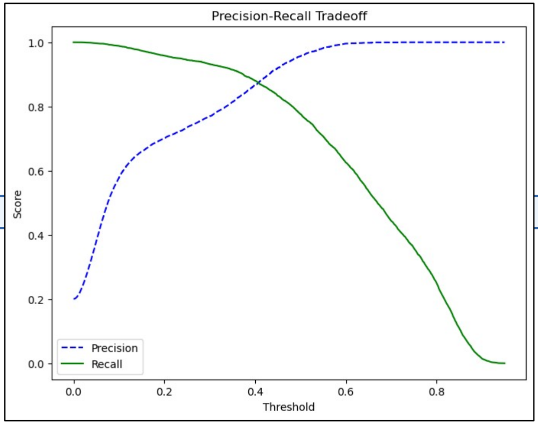
Prioritizing Low FN
or … *"Minimizing Bad Loans in Portfolio"*

Means fewer high-risk loans slip through as safe investments. However, this may increase FP, meaning more potentially profitable loans are wrongly rejected, leading to missed opportunities.

Prioritizing Low FP
or... *"Maximizing Investment Opportunities"*

Means more loans are considered investable, increasing potential returns. However, this raises the risk of FN errors, leading to a higher chance of investing in loans that default.

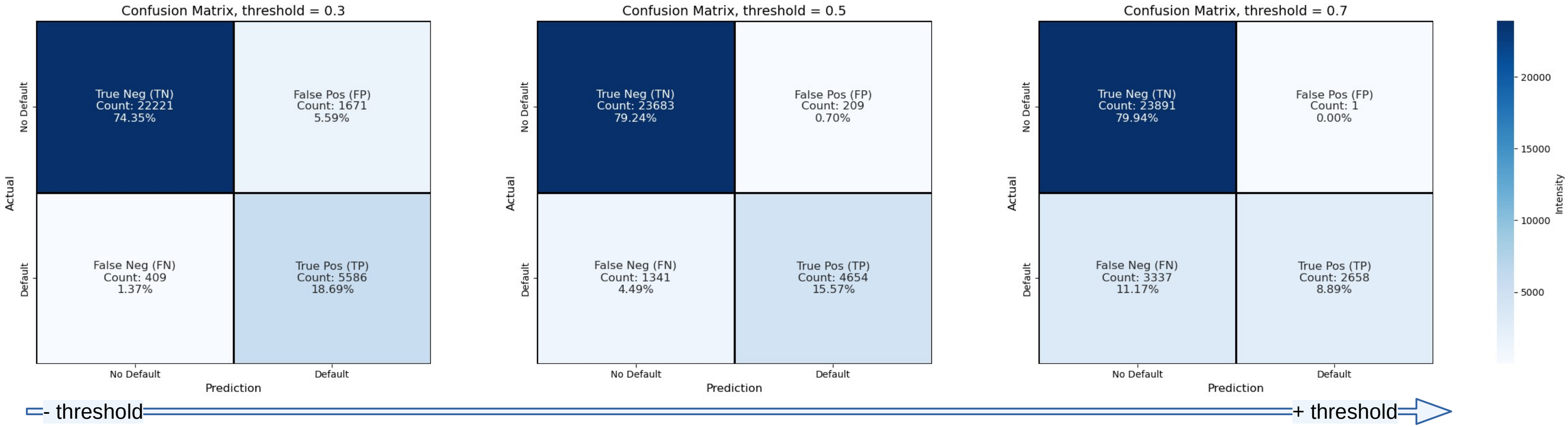
Adjusting the **classification threshold** allows control over the trade-off between false positives and false negatives...



Evaluation metrics were analyzed across **varying thresholds** to guide the trade-off selection

Tradeoff between FN and FP

False Negatives ⟸ trade-off between prioritizing optimization of FN vs. FP ⟹ **False Positives**

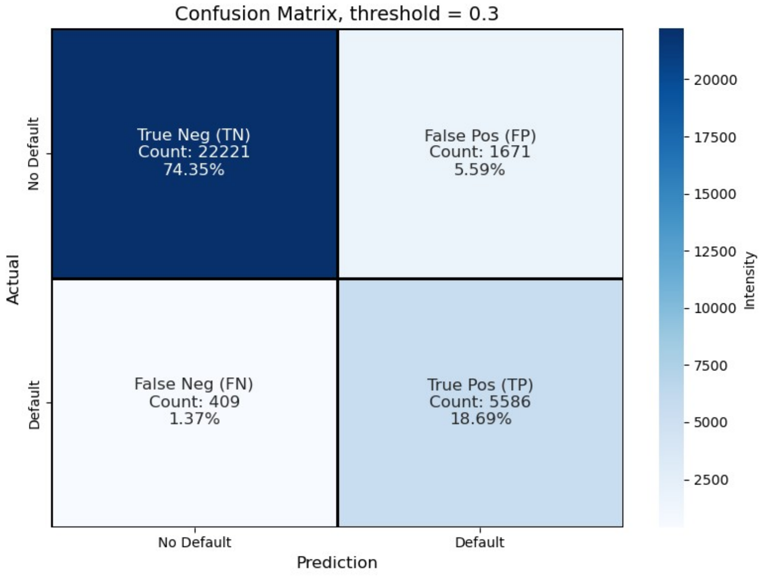


In line with our objective of analyzing investment strategies for the secondary market, **a threshold value of 0.3 was selected** to prioritize the **reduction of false negatives**. The rationale is that the cost of a false positive is relatively low in this context—given the abundance of investment opportunities in an open market, missing one is less critical than mistakenly investing in a loan that risks partial or total loss of principal.

Final Model Evaluation

The final model is a **Random Forest** with 250 estimators and a classification threshold set to 0.3. It is optimized to predict loan defaults with a focus on **minimizing false negatives**. On the test set, the model achieves a **recall of 0.93**, while maintaining a **low false positive rate**, with only 1.37% of instances incorrectly classified as defaults.

	precision	recall	f1-score	support
0	0.98	0.93	0.96	23892
1	0.77	0.93	0.84	5995
accuracy			0.93	29887
macro avg	0.88	0.93	0.90	29887
weighted avg	0.94	0.93	0.93	29887



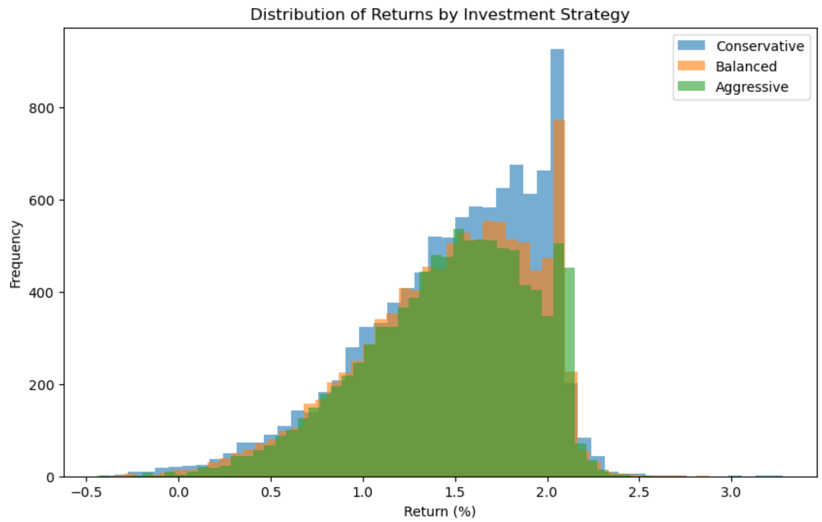
Assess investment strategies

Goal: What is the optimal investment strategy based on credit scores and interest rates?

Monte Carlo simulation with 10,000 iterations was performed on the open loans dataset, leveraging **model-predicted default probabilities** to estimate the profitability of each strategy.

Key components of the simulation included:

- Projection of remaining loan terms to model expected future cash flows from installment payments.
- Estimation of recovery rates for when a loan defaults, derived from historical performance data on closed loans.
- Application of Discounted Cash Flow (DCF) analysis to incorporate the time value of money into return calculations.
- Generation of summary metrics and visualizations to compare the distribution of return rates across different strategies.



The Monte Carlo simulation reveals that despite differing credit risk levels, the returns across Conservative, Balanced, and Aggressive strategies remain similar. This is due to Lending Club's efficient risk-adjusted pricing, where higher-risk loans offer higher interest but also higher default rates. Additionally, recovery rates and discounting mitigate extreme losses, while the secondary market filters out early defaults.

Future Improvements & Acknowledgments

Apply **resampling techniques** such as SMOTE or undersampling to address class imbalance and improve model sensitivity to minority classes (i.e., defaults).

Enhance model generalization by **revisiting feature selection** to identify and remove additional variables that may introduce subtle forms of data leakage, especially those only evident after model evaluation.

Explore **alternative algorithms** such as XGBoost, which can handle missing values and complex feature interactions more effectively, or Support Vector Classifiers (SVC) for comparison on smaller, well-preprocessed subsets.

Refine the investment simulation by **reviewing cost components**—such as transaction fees, taxes, or default recovery timelines—that may be misrepresented or double-counted, potentially deflating the estimated returns.